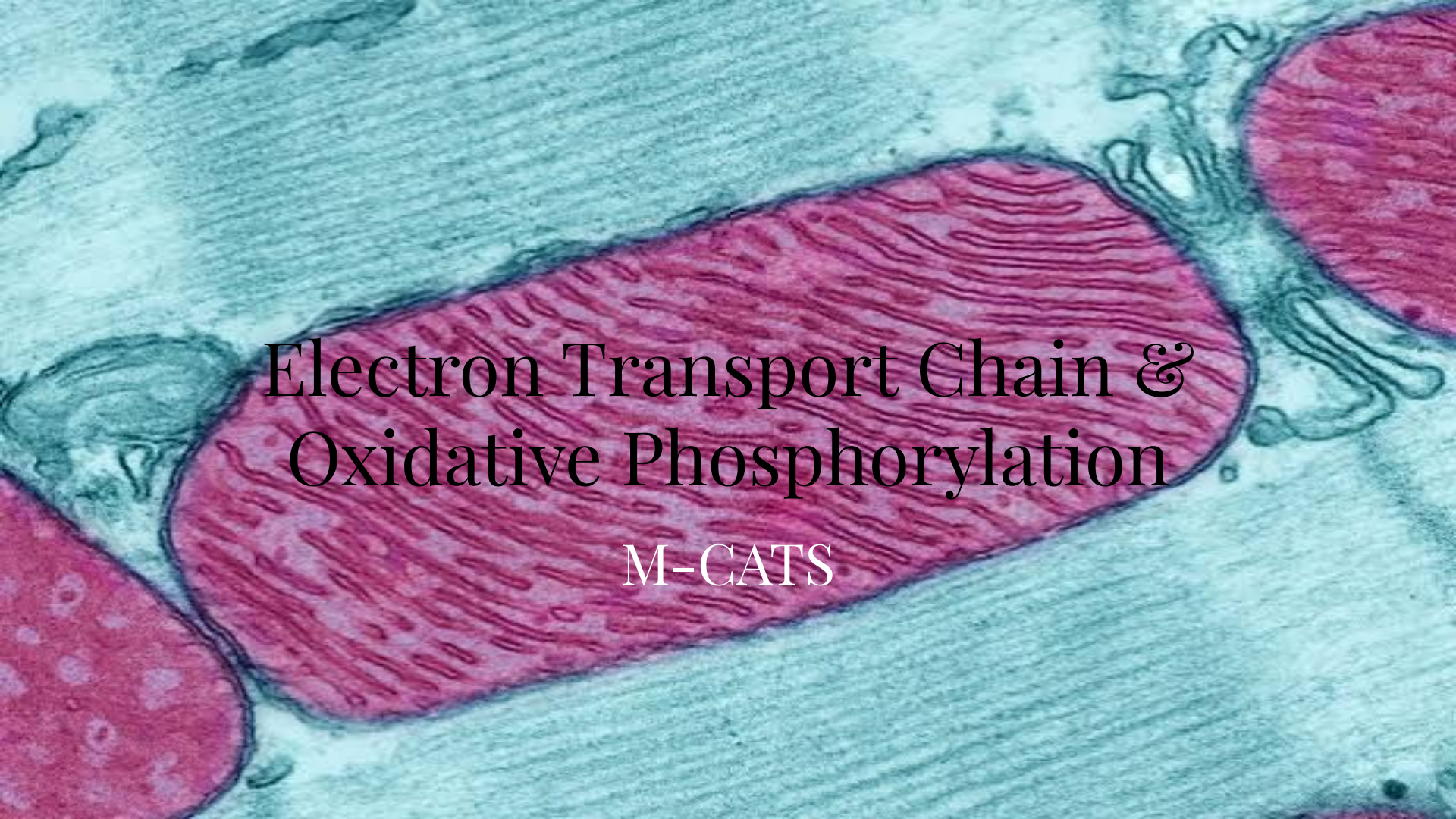


Electron Transport Chain & Oxidative Phosphorylation

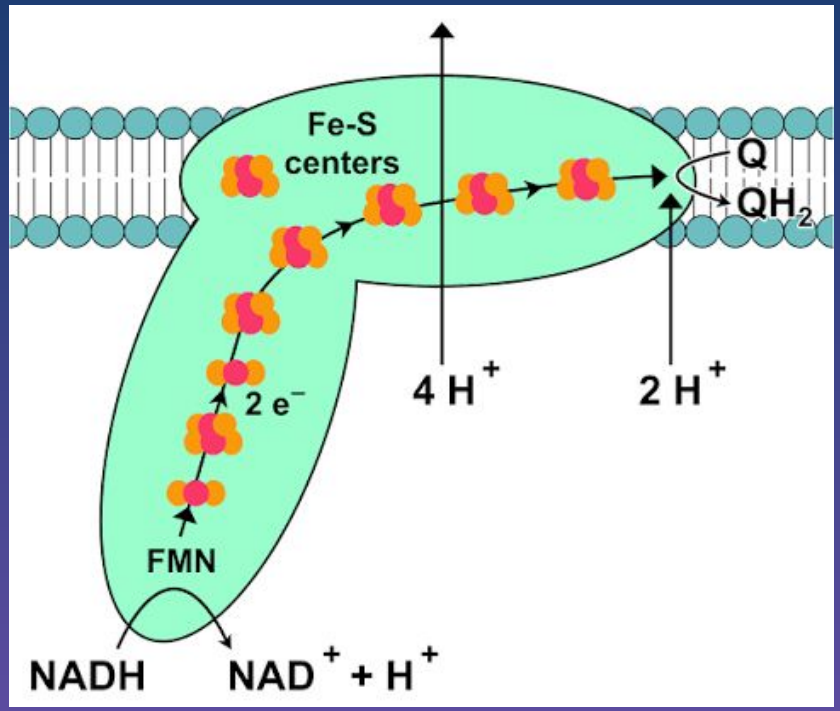
M-CATS

Complex 1 (NADH Dehydrogenase)

$NADH+H^+ + CoQ \twoheadrightarrow CoQH_2 + NAD^+$

$4\ H^+_{matrix} \twoheadrightarrow 4H^+_{intermembrane}$

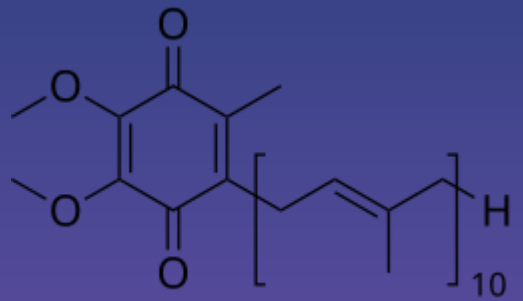
NADH ultimately powers the production of about 2.5 ATP



Wait, what is CoQ?

Coenzyme that acts as an intermediate electron carrier

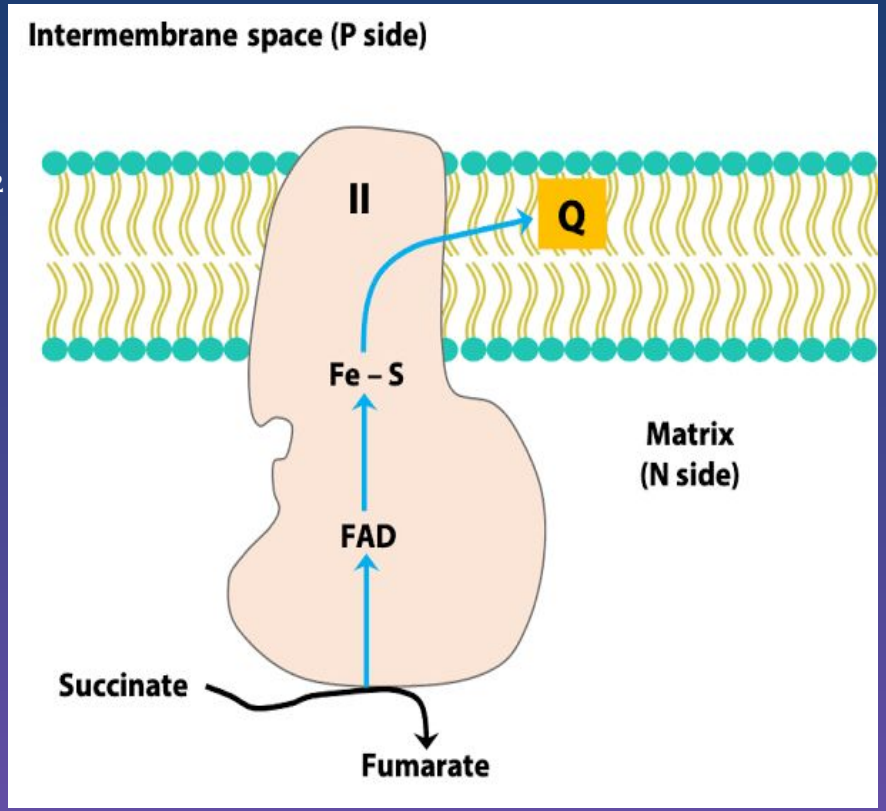
Ketone moieties become alcohols and ubiquinone becomes ubiquinol



Complex 2 (Succinate Dehydrogenase)

Succinate + CoQ + 2 H^+ ↠ Fumarate +$CoQH_2$

Because Complex 2 doesn't pump protons, it only produces 1.5 ATP per $FADH_2$



NADH Shuttles

NADH is produced during both anaerobic and aerobic respiration. However, NADH cannot cross the mitochondrial matrix to access Complex 1



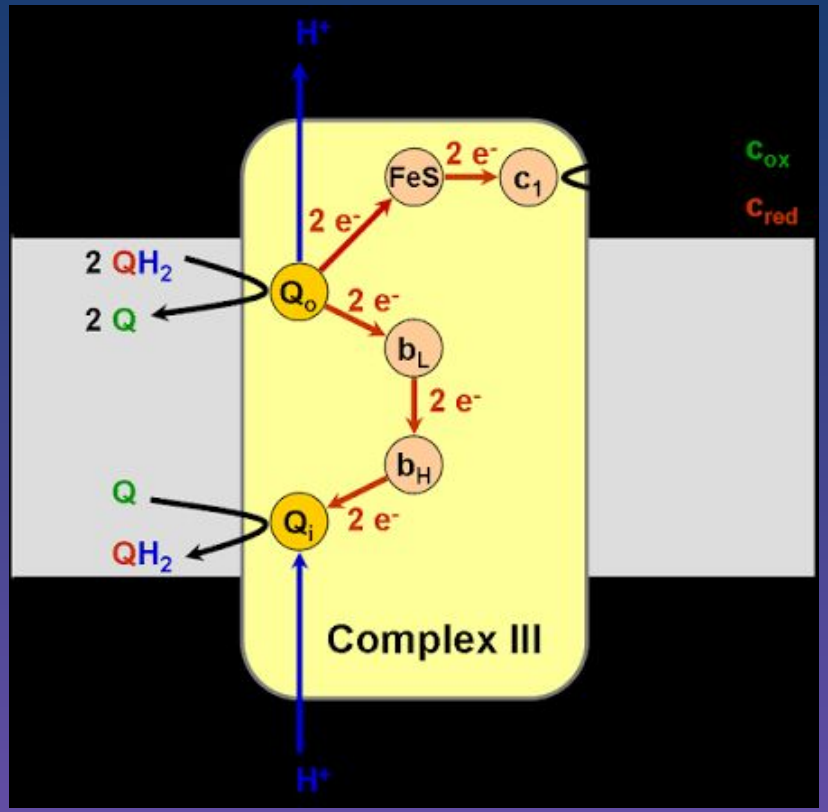
Shuttles:

- Glycerol 3-Phosphate
  - Uses NADH to form G3P from DHAP, then crosses membrane
  - FAD dependent G3P dehydrogenase forms $FADH_2$ from FAD
- Malate-Aspartate
  - NADH is used to turn cytosolic oxaloacetate into malate via malate dehydrogenase
  - Mitochondrial malate dehydrogenase then uses malate to form mitochondrial NADH
  - To return the malate to the cytosol, it is transaminated to aspartate from glutamate via aspartate transaminase

Complex 3 (Coenzyme Q - Cytochrome c reductase

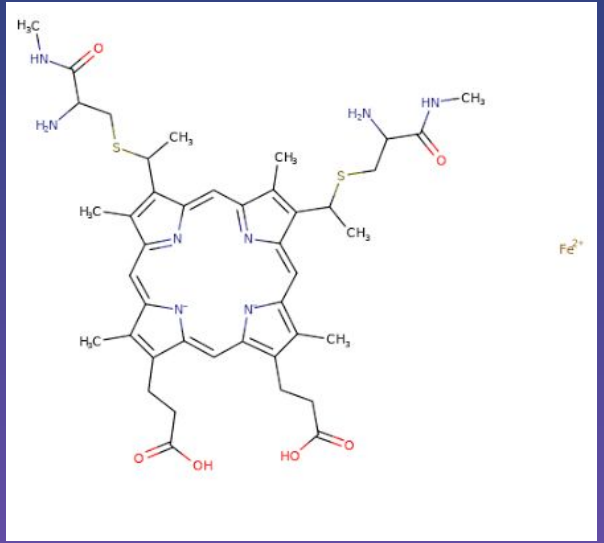
$CoQH_2 + 2cyt\ c_{Ox} \twoheadrightarrow CoQ + 2cyt\ c_{red}$

$4\ H^+_{matrix} \twoheadrightarrow 4H^+_{intermembrane}$



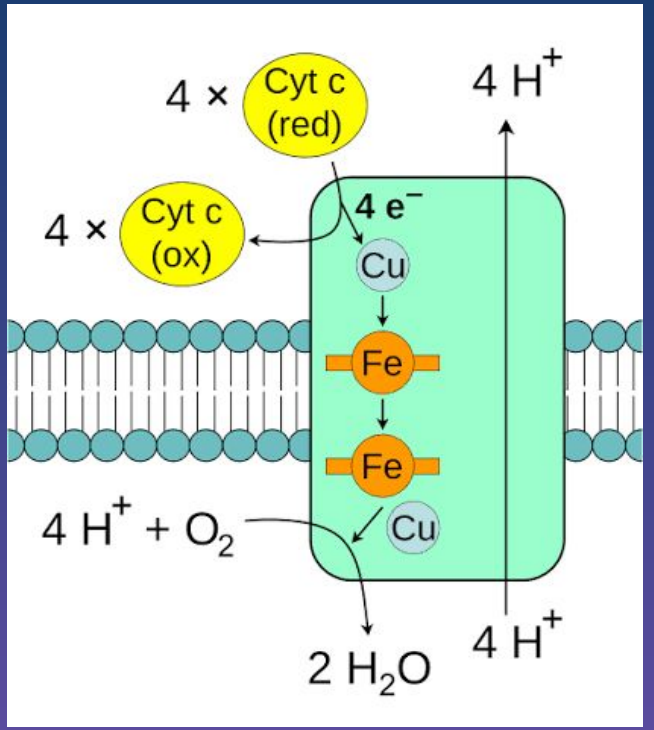
The Q cycle

- Cytochrome c can only carry one electron
- Complex 3 splits the electrons transferred from $CoQH_2$, ultimately forming another $CoQH_2$ while producing two reduced cyt c molecules



Complex 4 (Cytochrome C oxidase)

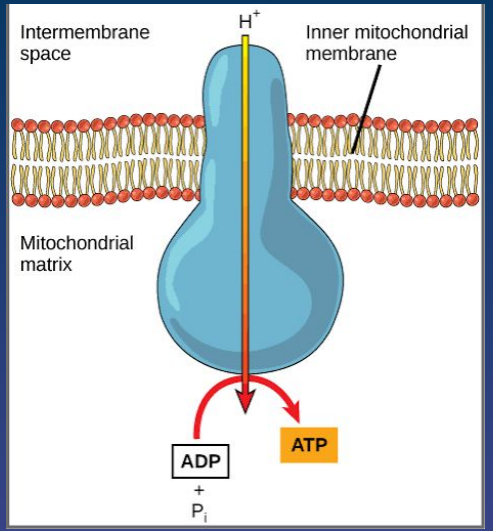
- Cytochrome a and a_3 are catalytic
- Cyanide is an inhibitor



$4\text{cyt } c_{red} + 4H^+ + O_2 \twoheadrightarrow 2H_2O + 4\text{cyt } c_{ox}$

$4\ H^+_{matrix} \twoheadrightarrow 4H^+_{intermembrane}$

Complex 5 (ATP Synthase)



Made up of several subunits

- F_o - proton channel
- F_1 - forms ATP from ADP
- Turbine structure
- Protons flow down the gradient (exergonic), so ATP can be synthesized (endergonic)

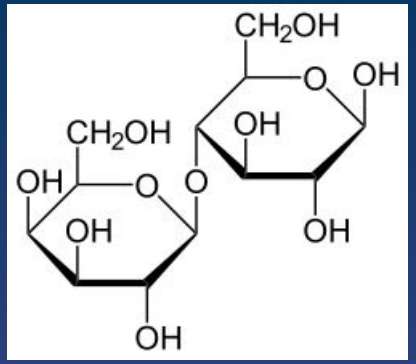
Regulation

- O_2 and ADP are required for this process and act as key regulators
- NADH accumulation during anaerobic conditions inhibits the TCA
- ADP also activates isocitrate dehydrogenase, increasing the rate of the citric acid cycle
- Products of the citric acid cycle in turn increase the rate of the ETC



Other Monosaccharides

Fructose and Galactose metabolism



- Sources of Fructose and Galactose:
- Lactase and Sucrase are present in the duodenum
- Galactose is phosphorylated by galactokinase and then converted into glucose 1 phosphate via galactose 1 phosphate uridyltransferase and an epimerase
- Fructose is phosphorylated in the liver by fructokinase, then joins the glycolysis pathway via aldolase B
- In other tissues, hexokinase phosphorylates fructose, but this is much slower

